Pharmacy Distribution ERP — Business Workflows & Operating Procedures

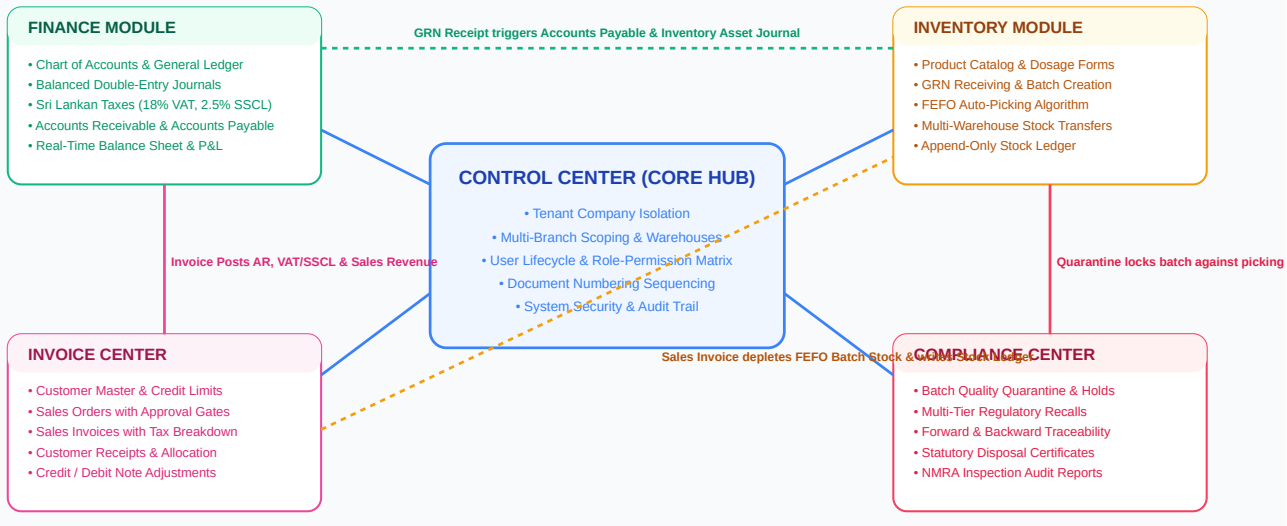
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Target Audience: Operations Directors, Warehouse Managers, Chief Accountants, Billing Officers, Quality Assurance Engineers
Version: 1.0.0 | **Publication Date:** 2026-10-15
Compliance Standards: WHO Good Distribution Practices (GDP), Sri Lankan NMRA Regulations, Inland Revenue Department (IRD) VAT/SSCL

1. Overview of Core Business Workflows

The Pharmacy Distribution ERP orchestrates five mission-critical business workflows: 1. **Procure-to-Pay (P2P):** Purchase Orders ➔ Physical GRN & Cold Chain Inspection ➔ Batch Creation ➔ Accounts Payable Posting. 2. **First-Expiry-First-Out (FEFO) Inventory Management:** Putaway ➔ Expiry Sorting ➔ Allocation Priority ➔ Internal Stock Transfers. 3. **Order-to-Cash (O2C):** Sales Orders ➔ Credit Validation ➔ FEFO Picking ➔ Sales Invoice ➔ Sri Lankan Dual-Tax Engine ➔ Dispatch. 4. **Automated Double-Entry Financial Posting:** Real-time Subledger to General Ledger Synchronization and Journal Balancing Invariant ($$\sum Dr = \sum Cr$$). 5. **Pharmaceutical Quarantine & Regulatory Recall:** Defective Lot Isolation ➔ Immediate Dispatch Blocking ➔ Customer Audit Trace ➔ Statutory NMRA Disposal.

Figure 10: Multi-Module Cross-Integration & Event Topology

Central Control Context, Real-Time Inventory-Finance Synchronization, Invoicing & Regulatory Quarantine



2. Procure-to-Pay (P2P) & Batch Receipt Workflow

2.1 Workflow Steps

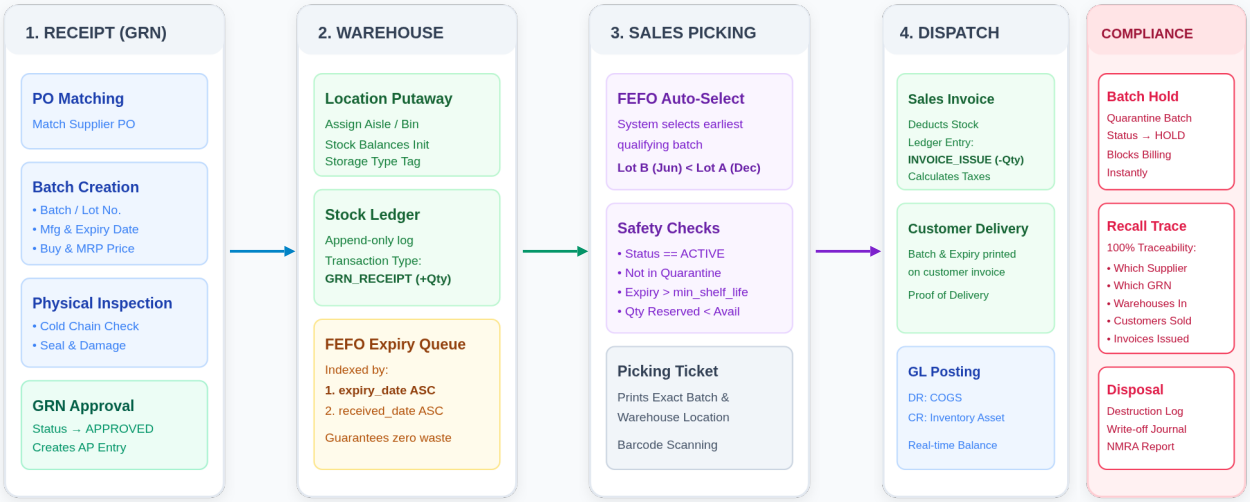
1. **Purchase Order Issuance:** Purchasing department creates an approved PO to the verified pharmaceutical manufacturer/supplier.
2. **Goods Delivery & Physical Inspection:** * Vehicle cold chain temperature dataloggers are verified ($$2^\circ ext{C} - 8^\circ ext{C}$$ for vaccines and biologics). * Outer packaging, tamper-evident seals, and NMRA registration numbers are inspected.
3. **Goods Receipt Note (GRN) Entry:** * Handled at `POST /api/v1/inventory/grns`. * For every product received, the warehouse officer inputs:
   - Manufacturer Batch / Lot Number (e.g. `LOT-B-202706`).
   - Manufacturing Date (`mfg_date`) and Expiration Date (`expiry_date`).
   - Received Quantity, Unit Purchase Price, and Statutory Maximum Retail Price (MRP).
4. **GRN Approval & State Transition:** * Status transitions from `DRAFT` ➔ `APPROVED`. * System initializes `product_batches` record with status `ACTIVE`. * Appends an immutable credit record to `stock_ledger_entries` with transaction type `GRN_RECEIPT`. * Automatically triggers Accounts Payable journal posting:
   - **DR:** `1200 - Pharmaceutical Inventory Asset`
   - **CR:** `2010 - Accounts Payable (Trade Suppliers)`

3. FEFO Inventory Management & Batch Expiry Control

Pharmaceutical regulations strictly forbid shipping medicines with less than the minimum statutory remaining shelf life (typically 6 months).

Figure 7: Pharmaceutical Batch Lifecycle & FEFO Dispatch Pipeline

Regulatory Traceability (First-Expiry-First-Out, Quarantine, Batch Recall & Audit Trail)



3.1 The FEFO Auto-Allocation Algorithm

When an order is placed, the backend query resolves the active batches for the product ordered using the following ordering predicate:

$$ ext{ORDER BY } exttt{expiry_date ASC}, exttt{created_at ASC}$$

```
SELECT id, batch_no, expiry_date, available_quantity, unit_price
FROM product_batches
WHERE product_id = ?
  AND company_id = ?
  AND status = 'ACTIVE'
  AND expiry_date > DATE_ADD(CURRENT_DATE, INTERVAL min_shelf_life_days DAY)
  AND (available_quantity - reserved_quantity) > 0
ORDER BY expiry_date ASC, id ASC
FOR UPDATE;
```

3.2 Concrete Allocation Example

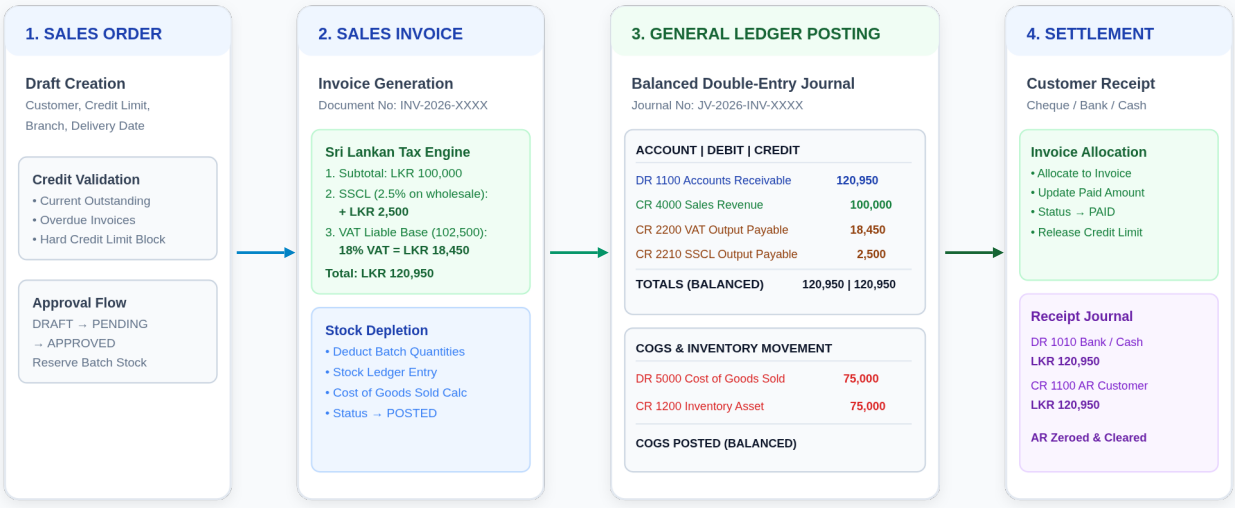
- **Batch A (Lot A):** Expiry Date: `2027-12-31` | Stock: 5,000 | Status: `ACTIVE`
- **Batch B (Lot B):** Expiry Date: `2027-06-30` | Stock: 1,500 | Status: `ACTIVE`
- **Incoming Customer Order:** 500 units of Amoxicillin 500mg.
- **Allocation Result: Batch B** is selected first because its expiry date (`2027-06-30`) precedes Batch A, preventing batch obsolescence and expiry waste.

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4. Order-to-Cash (O2C) & Sri Lankan Tax Calculation

Figure 8: Order-to-Cash & Automated Financial Posting Pipeline

End-to-End Sales Order, Sri Lankan Tax Engine (VAT 18% + SSCL 2.5%), Invoicing, Receipts & Balanced GL Posting



4.1 Sales Order & Credit Limit Enforcement

1. Billing officer selects customer (e.g. `Lanka Pharmacy (Pvt) Ltd`).
2. Backend inspects customer's current outstanding balance against approved credit limit: $$ ext{Available Credit} = ext{Credit Limit} - (ext{Outstanding Invoices} + ext{Pending Orders})$$
3. If order amount exceeds available credit, system blocks approval unless overridden by a Super Admin or Finance Manager with `control:credit_limit:override`.

4.2 Sri Lankan Dual-Tax Calculation Engine

Sri Lankan pharmaceutical wholesale transactions are governed by the Inland Revenue Act and statutory gazettes: 1. **SSCL (Social Security Contribution Levy):** Statutory rate of **2.50%** applied on the taxable turnover value. 2. **VAT (Value Added Tax):** Statutory rate of **18.00%** applied on the turnover including SSCL (unless the pharmaceutical item is gazetted as an exempt essential medicine under code `VAT_EXEMPT_PHARMA`).

Mathematical Tax Formulation

For a wholesale order with subtotal $$S$$:

$$ ext{SSCL Amount} = S imes 0.025$$

$$ ext{VAT Base} = S + ext{SSCL Amount} = S imes 1.025$$

$$ ext{VAT Amount} = egin{cases} 0.00 & ext{if product is VAT Exempt} \ ext{VAT Base} imes 0.18 & ext{if product is Standard VAT} \end{cases}$$

$$ ext{Grand Total} = S + ext{SSCL Amount} + ext{VAT Amount}$$

Real Calculation Example (Verified in Test Suite `FIN-014`)

- **Product Line Subtotal ($$S$$):** LKR 100,000.00
- **SSCL (2.50%):** LKR 2,500.00
- **VAT Base:** LKR 102,500.00
- **VAT (18.00% of LKR 102,500.00):** LKR 18,450.00
- **Invoice Grand Total: LKR 120,950.00**

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5. Automated Financial Journal Posting

When a Sales Invoice transitions to `POSTED`, the ERP executes an atomic transaction creating the balancing General Ledger entries:

5.1 Sales & Tax Revenue Journal

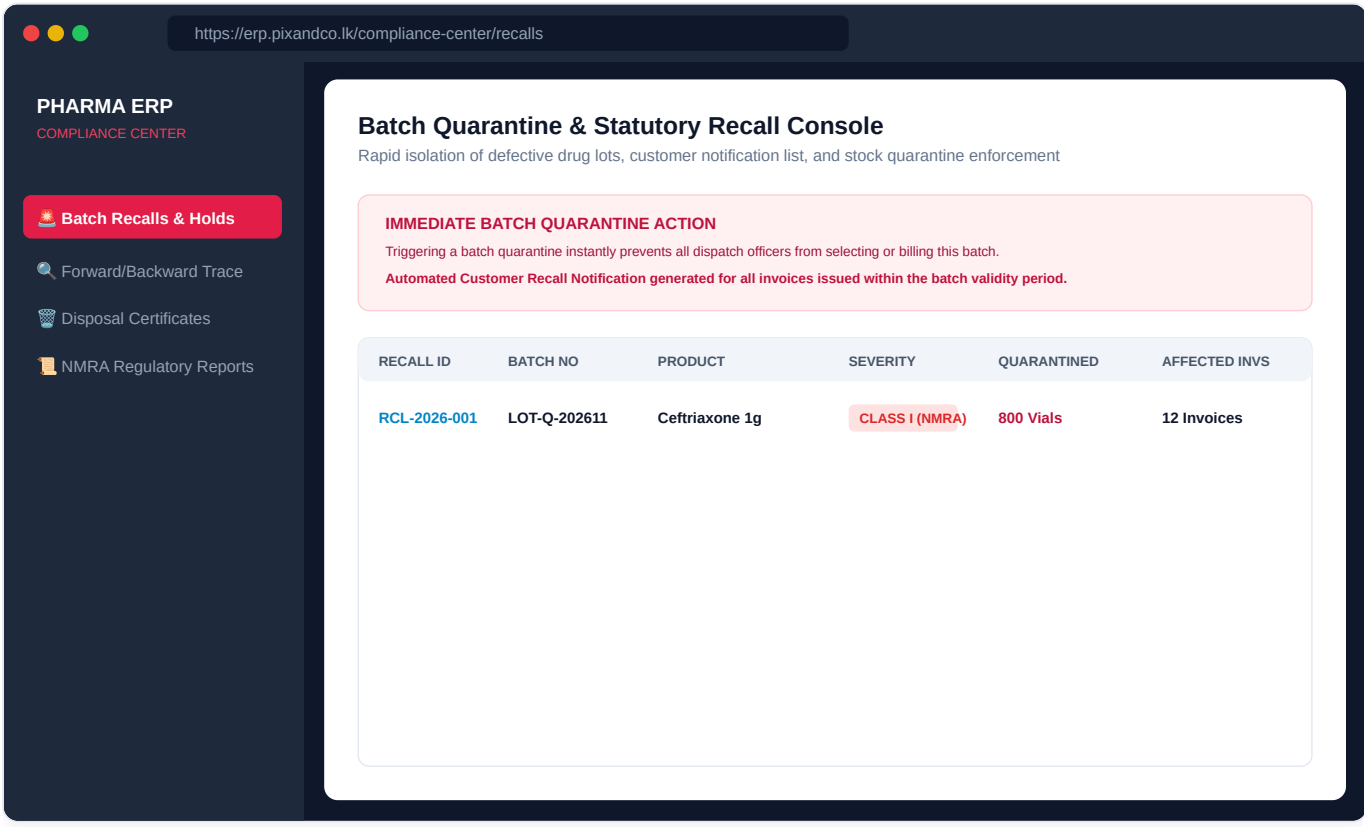
Account Code	Account Description	Debit (LKR)	Credit (LKR)
`1100`	Accounts Receivable (Lanka Pharmacy)	120,950.00	—
`4000`	Pharmaceutical Sales Revenue	—	100,000.00
`2200`	VAT Output Payable (18%)	—	18,450.00
`2210`	SSCL Output Payable (2.5%)	—	2,500.00
TOTALS	**BALANCED JOURNAL INVARIANT**	**120,950.00**	**120,950.00**

5.2 Cost of Goods Sold (COGS) & Inventory Depletion Journal

Concurrently, the system records the inventory asset reduction based on the batch purchase cost (e.g. LKR 75,000.00): | Account Code | Account Description | Debit (LKR) | Credit (LKR) | |---|---|---|---| | `5000` | Cost of Goods Sold (COGS) | 75,000.00 | — | | `1200` | Pharmaceutical Inventory Asset | — | 75,000.00 | | **TOTALS** | **BALANCED JOURNAL INVARIANT** | **75,000.00** | **75,000.00** |

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6. Pharmaceutical Batch Quarantine & Regulatory Recall Workflow



Under NMRA safety alerts or batch defect reports (e.g. microbial contamination or packaging defects), the Compliance Officer initiates an immediate quarantine:

6.1 Step 1: Immediate Batch Hold

1. Navigate to **Compliance Center ➔ Batch Holds**.
2. Select defective batch (e.g. `LOT-Q-202611`) and invoke `POST /api/v1/inventory/product-batches/:id/quarantine`.
3. System updates batch status from `ACTIVE` ➔ `HOLD`.
4. **Immediate Safeguard:** All billing selection queries automatically exclude `HOLD` batches. Any attempt to invoice this batch yields `HTTP 422 Unprocessable Entity ("Batch is quarantined")`.

6.2 Step 2: Customer Recall Audit Trace

1. The compliance engine queries `sales_invoice_items` joined with `sales_invoices` and `customers`: `sql SELECT c.name, c.phone, c.email, si.invoice_number, si.invoice_date, sii.quantity FROM sales_invoice_items sii JOIN sales_invoices si ON si.id = sii.sales_invoice_id JOIN customers c ON c.id = si.customer_id WHERE sii.batch_id = ? AND si.status = 'POSTED';`
2. System generates the **Statutory NMRA Recall Contact List** containing every pharmacy that received units of the quarantined batch.

6.3 Step 3: Disposal & Regulatory Write-Off

1. Once stock is returned from the market, stock is moved to the **Quarantine Store**.
2. A formal **Destruction Certificate** is recorded in Compliance Center.
3. Accounting writes off remaining inventory: - **DR:** `5100 - Inventory Loss / Damaged Goods Write-off` - **CR:** `1200 - Pharmaceutical Inventory Asset`